Table 1: **Vina score** of Top-100 diverse modes on LIT-PCBA. Mean ± standard deviations over 3 runs are reported.

	Method	GBA	IDH1	KAT2A	MAPK1	MTORC1
Reaction	SynFlowNet†	-9.27 ± 0.06	-10.40 ± 0.08	-9.41 ± 0.04	-8.92 ± 0.05	-10.84 ± 0.03
	RGFN†	-8.48 ± 0.06	-9.49 ± 0.13	-8.53 ± 0.11	-8.22 ± 0.15	-9.89 ± 0.06
	RxnFlow†	-9.62 ± 0.04	-10.95 ± 0.05	-9.73 ± 0.03	-9.30 ± 0.01	-11.39 ± 0.09
SMILES	S3-GFN	**-9.76** ± 0.08	**-11.43** ± 0.03	**-10.02** ± 0.10	**-9.38** ± 0.03	**-11.73** ± 0.01

	Method	OPRK1	PKM2	PPARG	TP53	VDR
Reaction	SynFlowNet†	-10.34 ± 0.07	-11.98 ± 0.12	-9.40 ± 0.05	-7.90 ± 0.10	-11.62 ± 0.13
	RGFN†	-9.61 ± 0.11	-10.96 ± 0.18	-8.53 ± 0.07	-7.07 ± 0.06	-10.86 ± 0.11
	RxnFlow†	-10.84 ± 0.03	-12.53 ± 0.02	-9.73 ± 0.02	-8.09 ± 0.06	-12.30 ± 0.07
SMILES	S3-GFN	**-11.48** ± 0.01	**-13.15** ± 0.03	**-9.89** ± 0.02	**-8.33** ± 0.10	**-12.86** ± 0.10

Table 2: **Results on sEH + QED.** Mean and standard deviations are reported with three independent runs.

	Positive (↑)	Pos. Top100 sEH (↑)	Pos. Top100 QED (↑)	Pos. Top100 Div. (↑)
Prior	0.299 ± 0.014	0.628 ± 0.008	0.710 ± 0.013	0.877 ± 0.001
S3-GFN (sEH)	0.945 ± 0.009	1.043 ± 0.001	0.664 ± 0.018	0.764 ± 0.000
S3-GFN (sEH + QED)	0.991 ± 0.002	1.001 ± 0.002	0.937 ± 0.002	0.759 ± 0.003

Table 3: **Performance under harsher constraint changes.** Mean ± standard deviation over 3 runs are reported.

	$\lvert\mathcal{R}\rvert = 15$ # positives in buffer: 2241			$\lvert\mathcal{R}\rvert = 10$ # positives in buffer: 1073		
	Zero-shot	RTB + RS	S3-GFN	Zero-shot	RTB + RS	S3-GFN
Avg. sEH (↑)	**1.010 ± 0.001**	0.943 ± 0.001	1.007 ± 0.001	**1.010 ± 0.001**	0.862 ± 0.042	1.007 ± 0.002
Positive Ratio (↑)	0.620 ± 0.023	**0.865 ± 0.020**	0.838 ± 0.012	0.174 ± 0.018	0.637 ± 0.061	**0.844 ± 0.010**
Num. Unique (↑)	**923.7 ± 3.1**	795.0 ± 8.5	861.3 ± 13.7	**923.7 ± 3.1**	550.7 ± 22.4	687.3 ± 14.4
Pos. Top100 sEH (↑)	1.035 ± 0.002	1.029 ± 0.001	**1.037 ± 0.000**	1.035 ± 0.002	1.016 ± 0.004	**1.038 ± 0.000**

Table 4: **Comparison on sEH under Enamine REAL reactions.** Mean ± standard deviation over 3 runs are reported.

Method	Positive (↑)	Pos. Top100 sEH (↑)	sEH (↑)	Diversity (↑)	SA (↓)	AiZynth (↑)	QED (↑)	Mol. weight (↓)
SFN (hb)	1.0	0.989 ± 0.005	0.907 ± 0.007	0.801 ± 0.022	2.876 ± 0.150	0.727 ± 0.255	0.694 ± 0.054	356.00 ± 6.31
S3-GFN (hb)	0.945 ± 0.009	1.043 ± 0.001	1.009 ± 0.000	0.764 ± 0.000	2.364 ± 0.016	0.990 ± 0.008	0.696 ± 0.007	350.35 ± 2.84
S3-GFN (REAL)	0.953 ± 0.009	1.054 ± 0.001	1.023 ± 0.002	0.757 ± 0.004	2.405 ± 0.013	0.980 ± 0.014	0.630 ± 0.002	372.99 ± 0.87